Supplementary Material: Mathematics or the Verifier?

Tan Sang Nguyen[1] and Duy Anh Tran[1]

Ton Duc Thang University, Ho Chi Minh City, Vietnam

A Related Work in Full

Self-training and its limits. STaR [19] introduced the generate–filter–retrain loop, and ReST [12], ReST-EM [11] and RFT [17] formalise it; ReST-EM, which like us restarts fine-tuning from the base model each iteration, reports diminishing test gains after the first round and a one-round regression on APPS. ReST filters machine-translation samples with a learned reward model and a rising threshold; the others filter by answer correctness. None of them measures how the filter's errors depend on the form of the answer, and they mostly use larger models than ours. Yue et al. [18] show that RL raises pass@1 while the base model wins at large pass@k, and Teacher-Free Self-Training [13] observes a pass@8 versus pass@64 crossover on a free-verifier domain, concluding that self-training "amplifies but does not compound". Training repeatedly on model-generated data erodes distribution tails [10], and Qiao et al. [8] argue, in theory and with experiments, that a selector seeing only a narrow slice of the target distribution accelerates collapse.

Verifier error. TinyV [16] measures that rule-based verifiers reject many correct solutions and repairs them with an LLM check. From Accuracy to Robustness [6] reports an average recall of 86% for rule-based verifiers, i.e. about 14% of correct responses rejected, but for `math-verify` itself on their MATH subset a recall of 0.95 (about 5% false negatives; their Table 5, on DeepSeek-R1-Distill-Qwen-32B outputs). Our MATH-500 audit gives 7.8% (95% Wilson interval 4.3–13.8%) on another model's solutions, a figure whose relation to the loop's own rate is not known (Section B), and a pipeline-specific defect alone rejects 15.5–18.3% of the loop's own solutions that a repaired verifier accepts, concentrated on about a tenth of the problems. These figures differ in verifier version, solution source and denominator, so we compare them only in order of magnitude. Imperfect Verifiers [1] gives the most formal noise model, with fixed false-accept and false-reject probabilities, and notes that real noise can depend on formatting but then assumes content-independent rates. Egashira et al. [4] classify verifier errors in RLVR on a synthetic arithmetic task (Qwen3-1.7B-Base and OLMo3-7B, with Qwen2.5-1.5B-Instruct, our model, in an appendix) with hand-designed patterns such as format-based false negatives, and find that systematic false negatives behave like random noise whereas false positives (rewards given to answers that are near the truth, or that contain a keyword) can cause plateau or collapse; they

also note that a domain whose queries are always rejected can at best plateau, which is the position of our unparseable-gold problems. They also track, per RL step, the false-negative rate on the policy's own rollouts and report that the policy first unlearns the behaviour that triggers false negatives; TinyV tracks the false-negative share of rollouts across GRPO steps (about a fifth per step in their runs). Our audit (Section 5.3) applies this measurement to offline SFT. Our setting differs in paradigm (offline rejection-sampling SFT with a fresh adapter per round, not on-policy RL) and in task (real benchmark problems). Our patch control cannot test their false-negative finding, because it changes too little of the training pool (Section F), and the much larger false-negative class we measured afterwards is present in both arms. Nor does our no-filter control test their false-positive finding: their false positives are systematic (a fixed wrong pattern that is always accepted), whereas no-filter admits every sample, about two thirds of them incorrect in unstructured ways, under SFT with a 3–4× larger training dose. It does add false positives in the plain sense, wrong answers in the training pool, but unstructured ones, and 62.4% of its round-0 pool is also truncated in training (against 24.5% for Run A). It did not end consistently worse than the filtered run, which we report only as a related observation. Their finding that systematic false negatives act like noise would, if it carries over, predict that a model need not learn to avoid a false-negative pattern, which is consistent with our audit, but our audit has too little power to test it.

Auditing self-improvement. Xu et al. [15] audit three rounds of rank-32 LoRA self-training of Qwen3-8B ($\eta = 10^{-4}$; filters by answer match or majority vote, plus a distillation arm from an external teacher as a positive control) against a frozen untrained control, and identify seven measurement failures that each invert a reported finding when its control is absent, including training-seed variance and a fixed token cap under style drift: their distillation arm, the most effective one, scored as the most destructive because its truncation rate rose from 16.5% to 65.3%. Self-training does not improve rarely-solved problems and corrupts problems solved at baseline. This is the study closest to ours in spirit. Our per-problem added/lost counts point the same way (more problems lost than gained), though without their frozen control and with net counts that only restate the change in pass@k (Section 3.4), and our adapters also drift in style (longer outputs, fewer `\boxed` answers), and a fixed token cap matters in our setting too: the untrained model already hits it in 41–48% of MATH-500 generations. Their finding is a warning that a decline under a fixed cap can be misread as lost capability; this is why we separate finished accuracy from termination and re-score at a larger cap (Section H). Their setup differs in model size, rank and filters; we lack their per-round frozen arm (we re-score the untrained model once, Section H) and a distillation arm, and we add a measured format-biased verifier, a learning-rate series and reference-solution and no-filter controls.

Reference-guided training. Kim et al. [7] show that self-distillation conditioned on the ground-truth solution can cut mathematical accuracy by up to 40% (Qwen3-1.7B/8B, DeepSeek-Distill-Qwen-7B and Olmo3-7B-Instruct) while

shortening responses, which they trace to suppressed epistemic verbalization. Our train-on-gold control is plain SFT on human reference solutions, not self-distillation, so it is at most a loose analogue; it shows a similar surface signature (shorter responses, lower accuracy) that may simply come from imitating terse references. We did not measure epistemic markers.

Positioning. In the literature we surveyed (searched through October 2026) we found no multi-round self-training study that uses a rule-based verifier with a measured format bias together with a matched patched-verifier control; the closest are Egashira et al. and Xu et al. Given that the patch turned out to be too weak to test anything (Section F), we do not present this combination as the contribution. The contribution is the absence of a detectable gain on the pre-set accuracy measures, with intervals and the smallest detectable effects, the description of what training at higher rates changes, and the design lessons. The per-round false-reject audit of Section 5.3 is not new: Egashira et al. track the false-negative rate on the policy's own rollouts per RL step and TinyV the false-negative share across GRPO steps. What we add is its application to offline SFT self-training, where a false reject only removes an example, and the conditions it needs there (conditioning on answer format and on the problems the defect affects, and a count of how many training examples could teach the model to avoid the defect).

B The Verifier in Full

Format probe. We re-rendered gold answers from GSM8K and MATH-500 in individually valid formats (bare number, trailing-zero decimal, `\frac`, `\dfrac`, `\tfrac`, boxed, dollar-delimited, sentence, with units) and asked the verifier to grade them against the original gold. On GSM8K plain rewrites are never rejected, while `\dfrac`/`\tfrac` are rejected 200/200 (the command is not parsed). On MATH-500, `\dfrac`/`\tfrac` are rejected 146/146 (100%), trailing-zero decimals 13/22, and boxed forms 41/200. Of those 41, 24 are gold answers our pipeline stores in forms the verifier cannot parse at all, so part of this figure measures our gold storage, not the verifier. The other 17 (17 of the 200 boxed probes, 8.5%) are compound answers such as $3\sqrt{13}$: the gold is stored without a delimiter, and `math-verify` applies its full LaTeX extraction only inside `\boxed{}` or `$...$`, so the bare gold is parsed to its first number and no longer matches a correct boxed prediction.

False-negative rate on real solutions. We took one public solution per problem for the first 200 problems of each dataset (teacher traces for GSM8K, Qwen2.5-7B-Instruct rollouts for MATH-500), not the 1.5B model's own outputs. Correctness was established by a scripted comparison of the boxed answer with the gold string after normalisation (`src/dacntt/read.py`: delimiters, braces, spaces and commas removed, `\dfrac`/`\tfrac` read as `\frac`, simple fractions compared by value), and we then counted how many of these correct solutions the verifier rejects. The false-negative rate is 7.8% on MATH-500 (10 of 128 correct solutions;

24 problems with unparseable gold excluded) and 0% on GSM8K (0/200; 95% Wilson interval 0–1.9%, one-sided upper bound 1.3%), identical for the `default` and `reward` extraction presets of `math-verify`. Of the ten MATH-500 cases, eight are rejected when the whole solution is verified but accepted when only the boxed answer is, which we labelled wrong-token extraction from a long solution, and two are rewrites the normalisation equates but the verifier does not match (e.g. `6 + 9i` for `6+9i`). Most of the eight are compound answers ($3\sqrt{13}$, $6 - 5i$, $11\sqrt{2}$, a list, a polynomial), so they overlap with the compound class of the format probe and very likely with the gold-storage defect below (verifying the boxed answer alone compares two bare strings, which the bare-gold parse matches); the label describes where the failure shows, not its cause. The 95% Wilson interval for 10/128 is 4.3–13.8%. The 7.8% estimates the false-negative rate on Qwen2.5-7B-Instruct solutions for problems whose gold the verifier can parse, among answers our normalisation recognises as correct. Correct answers in forms it does not equate are left out of the denominator, and the 24 problems with unparseable gold are counted as a separate defect below; relative to the loop's own rate, where the 1.5B model writes longer and, after training, less boxed solutions, the direction of the difference is not known. The audit also inspected only rejected solutions, so it cannot measure the false-positive rate; that the verifier rarely accepts wrong answers is an assumption here, not a measurement.

Unparseable gold in the loop. Over all 500 MATH-500 problems the verifier cannot parse 51 stored gold answers (10.2%); 11–18 of them fall in each 150-problem exam partition. The loop keeps these problems: every sample is rejected, so they never enter the training pool and always count as unsolved. They lower absolute pass rates by up to 12 points but are constant within a partition, so they cancel in within-run and same-partition differences; they do not cancel in ratios or in levels compared across partitions. Wrapping the gold in `\boxed{}`, the re-score described below, makes 46 of these 51 golds parseable, so the two corrections overlap and should not be added (`experiments/viec3/review_cpu/dot4/unparsed.py`).

The patch. For the control we repaired the most *severe* identified failure mode with a normalisation pass that rewrites `\dfrac` and `\tfrac` to `\frac` before parsing, removing the 100%-rejection class and nothing else. The patched verifier is therefore still biased: it keeps the eight audited "wrong-token" false negatives (most of them probably the gold-storage defect) and the gold-storage defect itself.

The gold-storage defect in the loop's own outputs (post hoc). After the runs we re-scored the saved generations with a second normalisation that wraps the gold in `\boxed{}` when the prediction contains one and the gold does not (`experiments/viec3/review_cpu/`: `patch_dose.py`, `dose.py`, `flips.py`). This defect is far more frequent than the `\dfrac` class. Over all 500 problems it rejects 15.5–18.3% of the generations the re-scored verifier accepts (Run A at 5×10^{-4}, seeds 1 and 2, and at 10^{-5}, seed 0, every round). The kept pool covers only the 350 selection problems, where the share is about the same (17.8–18.8% at

round 0): accepting these generations would enlarge the round-0 pool by about a fifth (Run A at seed 1: 440 → 542, +23%; at 10^{-5}: 872 → 1,066, +22%), against 0–4 examples (under 1%) for the `\dfrac` patch. In a hand check of 30 random flips, 29 pairs were the same value written differently (e.g. `\frac43` and `\frac{4}{3}`; one prediction had an empty box). The 15.5–18.3% assumes that the re-scored verifier itself accepts no wrong answer, which we did not measure. On the exam set the defect lowers absolute pass@1 by 6.7–8.0 points at round 0 and throughout the 10^{-5} runs, by 4.7 points at round 2 under 5×10^{-4}, and by 0.7–3.3 points for train-on-gold after round 0. The absolute offset is therefore not constant. As a share of re-scored exam pass@1 it is 6.7/40.0 and 4.7/28.0 (both 17%) for Run A at seed 1, 6.7/38.0 = 17.6% and 4.7/32.0 = 14.7% at seed 2, and about 20% throughout at 10^{-5}. Over all 500 problems and all samples (the r_t of Section 5.3, a different denominator) it falls from 18.3 to 15.7% and from 18.1 to 15.5% at 5×10^{-4} and from 18.3 to 16.8% at 10^{-5}; Section 5.3 shows that this fall follows the falling `\boxed` share and that the defect stays at 94–98% on the problems where it fires. With the original verifier the declines below are *understated*: the original-verifier sample-accuracy changes are 0.73–0.78 of the re-scored ones in the three self-training runs and 0.80–0.85 in train-on-gold. These ratios are observations for five runs, not a correction factor: intervals do not scale with them, so we do not rescale the other runs or the 2.2-point bound. We keep the original verifier's numbers in every table, since that is the verifier the loop trained with; they are therefore 4.7–8.0 points too low in absolute terms for self-training. Re-scored, with paired bootstrap intervals as in Section 3.4, sample accuracy changes by -10.7 $[-14.8, -6.7]$ and -6.0 $[-9.7, -2.2]$ points for Run A at 5×10^{-4} (seeds 1 and 2; original verifier -8.3 and -4.7), by -3.3 $[-5.9, -0.7]$ for Run A at 10^{-5}, seed 0 (original -2.4), and by -24.7 and -22.0 for train-on-gold. In pass@1 the same runs go from 40.0% to 28.0% and from 38.0% to 32.0% (interval $[-14.0, +2.0]$ for the latter), from 40.0% to 36.7%, and from 40.0% to 16.0% and from 38.0% to 13.3%. Only these five runs were re-scored, three of them self-training; that the other runs' declines would also survive a re-score is an extrapolation from these three. Both arms of the patch control (Section F) carry this defect.

C Uncertainty in Full

Uncertainty. For every within-run change (round 0 against the last round) and every paired comparison of two runs on the same exam partition we report a 95% interval from a paired problem-level bootstrap: we resample the 150 exam problems with replacement (10,000 times), recompute the statistic for both rounds or runs on the same resampled problems, and take the 2.5th and 97.5th percentiles. Resampling problems rather than samples respects the clustering of the k samples within a problem. The per-sample correctness was recomputed from the saved generations with the same verifier; it reproduces the reported pass rates except for six round values that differ by one problem (0.6–0.7 points), most likely because of `math-verify` comparison timeouts

Table S1. The two designs. They differ in more than the intended variables; the most quantifiable unintended difference is the training dose per round (about 4.5× as many optimizer steps in Design P).

	Design P (pooled)	Design I (isolated)
Datasets per run	GSM8K + MATH-500	one only
Adapter	one, shared	one per dataset
Training mixture	≈ 69% GSM8K	single-domain
Evaluation items	same as selection	held-out 30%
Training examples, round 0 (1.5B)	3,962	872 (A), 1,901 (C)
Verifier control	none	Run B (patched)
Models	0.5B and 1.5B	1.5B

(`experiments/viec3/bootstrap_ci.py`). The post hoc measures of Section H (Table S7, the matched column of Table 2 and the decomposition) use a separate re-scoring of the same generations (`review_cpu/scored/`), which differs from the values behind Tables 1 and 2 by up to one problem in some further rounds (e.g. 10^{-4}, seed 1, round 2: pass@1 33.3 against 34.0%); no conclusion depends on these differences. These intervals cover the choice of problems and the sampling of generations for a given trained adapter; they do *not* cover training-seed variance, which we can only see informally across our two or three seeds, so they understate the uncertainty of run-level conclusions. Our own data suggest by how much: a replicate of Run A at 5×10^{-4}, seed 1, with the same configuration, ends 3.2 points of sample accuracy away from the original (Section H); and at seed 2 Runs A and B, whose round-0 training pools are identical (424 examples each), end 7.3 points apart in round-2 pass@1 (interval $[-14.0, -0.7]$, Section G), though that is a single interval near zero. A single interval that excludes zero is therefore not sufficient evidence of a reproducible effect, and one that includes zero does not rule an effect out. We also make no correction for multiplicity: Table 2 alone holds 76 intervals and the text about ten more. If no change were real, about four of the 76 would exclude zero by chance; many changes in the table are clearly real (the train-on-gold and GSM8K declines), and the intervals are not independent, so this is not a test that any particular interval is a false alarm, only a reason to rest conclusions on patterns across runs and seeds rather than on single intervals. Our only estimate of training-seed variance rests on one same-configuration pair, so it is itself very imprecise. Single proportions use Wilson intervals.

D Design P in Full

Table S3 reports pass@1 and pass@8 for both models over five rounds of the pooled design, as computed during the run. A later recomputation (Table S2) reproduces every pooled value with both verifier presets except round-3 pass@8, where Table S3's 63.6% matches the `default` preset and the `reward` preset used for selection gives 63.5%. The 1.5B decline is severe and accelerating; the

Table S2. Design P, 1.5B, split by dataset, recomputed from the saved generations (preset `reward`, the one used for selection; the `default` preset differs only at MATH-500 round-3 pass@8, 45.0%). Pooled, the recomputed round-3 pass@8 is 63.5% (`reward`) and 63.6% (`default`), against 63.6% in Table S3.

Dataset	Metric	R0	R1	R2	R3	R4	Δ
GSM8K	pass@1	69.6%	62.0%	61.0%	45.6%	26.6%	−43.0 pt
GSM8K	pass@8	91.4%	90.4%	89.0%	82.2%	65.8%	−25.6 pt
MATH-500	pass@1	30.4%	30.4%	28.0%	19.2%	10.6%	−19.8 pt
MATH-500	pass@8	53.8%	52.6%	51.4%	44.8%	31.4%	−22.4 pt

0.5B decline is small (−3.7 and −5.6 points). Splitting the 1.5B run by dataset (Table S2) shows both datasets falling heavily, although GSM8K has a measured 0% false-negative rate, though that rate was measured on external solutions in the untrained model's style, not on Design P's late-round outputs. The accepted pool also contracts every round (1.5B: 3,962, 3,730, 3,395, 2,638, 1,566; 0.5B: 2,533 to 2,123; `experiments/viec3/review_cpu/dot4/poolP.py`) and the 1.5B output style shifts, with `\boxed{}` falling from 5,608 to 2,693 of 8,000 generations while dollar-delimited math rises from 1,453 to 3,406.

Design P cannot be interpreted causally, for three reasons. First, each round trains one adapter on the union of both datasets' accepted solutions, of which GSM8K supplies 68.4–70.2% in every round, so either dataset's trajectory is a joint function of both pools. Second, it scores each round on the problems it also used to select training data. We expected this to *raise* later-round scores, if anything. Design I lets us check that expectation, since it also generates on its 350 selection problems: there, Run A's sample accuracy on the selection problems falls as much as or more than on the exam at the two higher rates (e.g. Run A at 5×10^{-4}: −8.2 and −5.3 points against −8.3 and −4.7 on the exam; at 10^{-4}, seed 1: −6.2 against −1.5; no intervals); at 10^{-5} it falls slightly less (−1.0 and −1.1 against −2.4 and −1.5), and Run C's selection problems fall less at 5×10^{-4} (−11.3 against −15.3) and slightly more at 10^{-5} (−13.8 against −13.1; exam values from Table 2). Training on a problem's own accepted solutions therefore did not protect it, so selection-set scoring is unlikely to have masked a decline; since the selection problems sometimes fall more, it may have enlarged one, and it remains a candidate contributor. Third, and not anticipated when we designed the follow-up, Design P trains on far more examples per round: 3,962 at round 0 for the 1.5B model, against 872 for the isolated MATH-500 run at the same learning rate. With one epoch and the same batch size this is about 4.5× as many optimizer steps (Design P's training logs were not saved, so this is inferred from pool size). Section G suggests, on the evidence of one seed, that a higher learning rate produces more decline, with a falling `\boxed` share as here (pass@1 falls faster than pass@k in only two of its four runs at 5×10^{-4}), and Run C, which takes about twice Run A's steps per round at the same rate, is the one 10^{-5} run that declines clearly. One reading consistent with these data is that Design P's adapter changed the model more. It rests

on two assumptions we have not tested: that more optimizer steps act like a higher learning rate, and that Run C's decline comes from its dose rather than its dataset (a single comparator). The time course adds a constraint: with a fresh adapter each round, the round-t model is trained on the round-$(t-1)$ pool alone, so the restart-from-base design leaves the data as the only channel from one round to the next. The pools shrink, yet the largest pass@1 drops (-12.1 and -13.8 points) come from the two smallest pools (3,395 and 2,638), so dose per round cannot by itself explain the acceleration. Dose and data need not be alternatives, though: a large dose in an early round can change what the model generates, and so what the next pool contains, which is one way dose could act through the data. The 0.5B model, trained on 2,533 examples at round 0 (about $2.9\times$ Run A), declines only a little (-3.7 points of pass@1), which weighs against dose alone, though it is a different model with a smaller pool. Pooling or scoring may also have contributed; no run separates these explanations.

Table S3. Design P: pass@1 and pass@8 over five GVT rounds, pooled GSM8K + MATH-500, scored on the selection set. Round 0 is the untrained base model. Section 4.1 explains why these numbers cannot be attributed to any single cause.

Model	Metric	R0	R1	R2	R3	R4	Δ (R0→R4)
0.5B	pass@1	31.1%	31.9%	29.9%	28.9%	27.4%	-3.7 pt
0.5B	pass@8	60.5%	58.9%	56.1%	55.5%	54.9%	-5.6 pt
1.5B	pass@1	50.0%	46.2%	44.5%	32.4%	18.6%	$\mathbf{-31.4}$ pt
1.5B	pass@8	72.6%	71.5%	70.2%	63.6%	48.6%	$\mathbf{-24.0}$ pt

E Design I at 10^{-5} in Full

Table S5 reports the isolated runs on their held-out exam sets. pass@8 ends within 4 points of its round-0 value in every run. On MATH-500, pass@1 is 2.7–5.3 points lower at round 1 than at round 0 and roughly flat afterwards, and 2.7–4.0 points lower at round 4; the bootstrap intervals for these round-0-to-round-4 changes are between $[-10.7, +2.7]$ and $[-9.3, +4.0]$, all including zero (Table 2). With one first-sample draw per problem per round, a step at round 1 cannot be distinguished from noise around a flat or slightly negative trend, so we claim only that there is no evidence of compounding on MATH-500 at this rate (Run C, below, loses most of its pass@1 in its last round), not that the first round has a one-off cost. The less noisy measures do move: sample accuracy falls by 1.2–3.2 points in these four runs, with one interval (B, seed 0) excluding zero, and the (post hoc) matched finished accuracy falls by 1.1–4.2 points, with intervals excluding zero at seed 0 for both A and B. These seed-0 intervals only just exclude zero (upper ends -0.6 and 0.0 for sample accuracy, -0.5 and -0.9 matched), A and B at seed 0 share their round-0 generations and almost all of their first pool, so they are close to one replicate, and at seed 1 nothing excludes

zero. Because these are near-replicate intervals that only just exclude zero, we read this as a possible small decline at this rate, not an established one; what we do not see is a gain of the size the design could detect. Run C (GSM8K) declines more: pass@1 falls 12.7 points, 9.3 of them in the final round (63.3% → 54.0%; interval for the whole change $[-21.3, -4.0]$), and sample accuracy 13.1 points, while pass@8 falls 2.7 points (interval including zero) and the cap-hit share rises only from 3.5% to 4.2%. Finished accuracy falls 12.8 points matched, and the decomposition of Section H puts 97% of the sample-accuracy decline in finished accuracy, so this decline is not a termination effect. It is a single run, and it took about twice as many optimizer steps per round as Run A (951 against 436 at round 0), so we do not attribute it to the dataset alone.

How much training happened at 10^{-5}*.* The training loss does fall at this rate: within round 0 of Run A it goes from 0.72 to 0.38 (seed 0) and from 0.70 to 0.40 (seed 1) over about 420 steps, with an epoch mean of 0.47, against epoch means of 0.24–0.30 at 10^{-4}–5×10^{-4}. The adapter therefore learns, but slowly, and it changes the outputs little: mean length and the share of generations that hit the token cap are essentially unchanged from round 0 to round 4 (cap-hit 46.9 → 47.4% at seed 0 and 40.7 → 41.8% at seed 1 for Run A; Table S7). This holds for the MATH-500 runs at about 420–436 steps per round; Run C at the same rate, with about twice the steps and a different dataset, does change its outputs. With that one comparison we cannot tell whether "little change" depends on the steps, the dataset or both. Length and cap-hit share are coarse statistics, and we did not measure finer ones (e.g. per-token divergence from the base model), so the evidence that the adapter did little is weak. A flat pass@k curve in the MATH-500 runs is therefore not evidence that GVT is stable.

Noise. Round 0 is the untrained base model, so round-0 differences between seeds mostly measure how hard each partition is: round-0 pass@1 is 32.0% at seed 0 and 36.7% at seed 1, and on the third partition used later round-0 pass@4 is 40.0% against 50.0% and 51.3% on the other two (Table 1). These levels cancel in within-run changes. Noise in the changes themselves is better seen elsewhere: re-sampling the same model on the same problems moves pass@1 by up to 6.7 points (Section H), and a single 150-problem pass@1 has a standard error of about 3.8 points. Agreement across seeds tests partition and training variance together; training variance alone is seen only in same-partition pairs (A against B at the same seed, Section G, and the replicate of Section H). We therefore report the bootstrap intervals above rather than judge differences against one "noise floor".

F Patch Control and Answer Diversity in Full

Runs A and B are identical in dataset, seed, split, model and training configuration and differ only in whether `\dfrac`/`\tfrac` are normalised before verification. Their curves are close (Table S5): at round 4, B minus A is −0.7 [−3.3, +2.0]

Table S4. All runs (Qwen2.5-1.5B-Instruct unless stated). Design P was run once per model. "Pool" is the round-0 kept pool (training examples; optimizer steps are half of it).

Design / run	Data	Rate	k, rounds	Seeds	Pool
P (0.5B, 1.5B)	GSM8K+MATH-500	10^{-5}	8, 5	one each	2,533, 3,962
I: A, B	MATH-500	10^{-5}	8, 5	0, 1	872, 833 (A)
I: C	GSM8K	10^{-5}	8, 5	0	1,901
I: A	MATH-500	10^{-4}	4, 3	1, 2, 3	424–455
I: A	MATH-500	2×10^{-4}	4, 3	1, 2	440, 424
I: A, B	MATH-500	5×10^{-4}	4, 3	1, 2	440, 424
I: no filter, gold	MATH-500	5×10^{-4}	4, 3	1, 2	1,400, 350
I: A replicate, 1024 limit	MATH-500	5×10^{-4}	4, 3	1	440
I: C	GSM8K	5×10^{-4}	4, 3	1	971

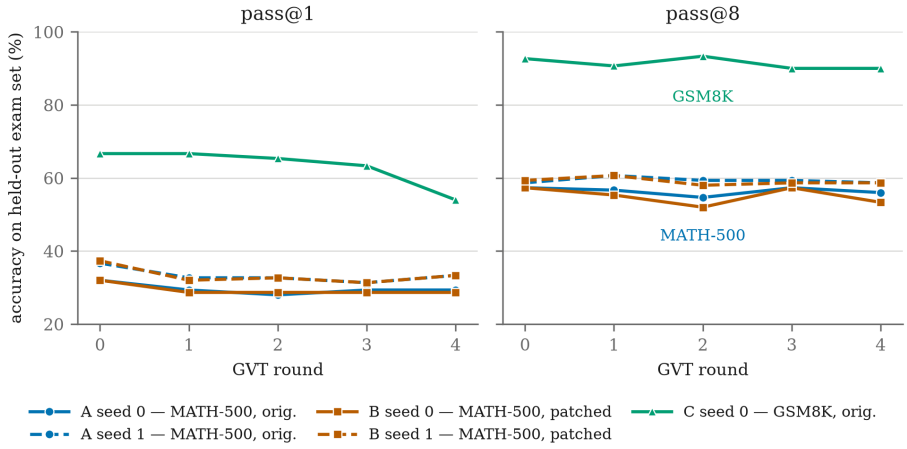


Fig. S1. Design I at 10^{-5}: pass@1 (left) and pass@8 (right) on the held-out exam set, by round. Colour identifies the run, line style the seed, marker the run again, so the figure survives greyscale printing. Runs A and B track each other closely at both seeds. pass@8 stays within a few points in every run. The one large visible movement is Run C's pass@1 in the final round.

points of pass@1 at seed 0 and +0.0 [−2.7, +2.7] at seed 1 (95% intervals). Run B's exam is graded with the patched verifier, so these differences also contain a grading offset: at seed 1 the identical round-0 generations already score 0.6 points higher in B, so the model part of the seed-1 difference is about −0.6. Narrow intervals here bound the difference between two nearly identical trajectories, not a patch effect. This null is uninformative, for two reasons we should have checked before running it. First, the treatment is tiny: `\dfrac` appears in 20–29 of the 4,000 generations per round (0.50–0.73%), `\tfrac` never, and only the correct ones among those are wrongly discarded; the kept pools of A and B

Table S5. Design I: pass@1 and pass@8 on the held-out exam set (150 problems), Qwen2.5-1.5B, learning rate 10^{-5}. Runs A and B differ only in the `\dfrac/\tfrac` patch. Each seed draws its own exam partition, so round-0 values differ between seeds.

Run	Metric	R0	R1	R2	R3	R4	Δ
A, seed 0 (MATH-500, orig.)	pass@1	32.0%	29.3%	28.0%	29.3%	29.3%	−2.7 pt
A, seed 0 (MATH-500, orig.)	pass@8	57.3%	56.7%	54.7%	57.3%	56.0%	−1.3 pt
A, seed 1 (MATH-500, orig.)	pass@1	36.7%	32.7%	32.7%	31.3%	33.3%	−3.3 pt
A, seed 1 (MATH-500, orig.)	pass@8	58.7%	60.7%	59.3%	59.3%	58.7%	0.0 pt
B, seed 0 (MATH-500, patched)	pass@1	32.0%	28.7%	28.7%	28.7%	28.7%	−3.3 pt
B, seed 0 (MATH-500, patched)	pass@8	57.3%	55.3%	52.0%	57.3%	53.3%	−4.0 pt
B, seed 1 (MATH-500, patched)	pass@1	37.3%	32.0%	32.7%	31.3%	33.3%	−4.0 pt
B, seed 1 (MATH-500, patched)	pass@8	59.3%	60.7%	58.0%	58.7%	58.7%	−0.7 pt
C, seed 0 (GSM8K, orig.)	pass@1	66.7%	66.7%	65.3%	63.3%	54.0%	−12.7 pt
C, seed 0 (GSM8K, orig.)	pass@8	92.7%	90.7%	93.3%	90.0%	90.0%	−2.7 pt

differ by 4 examples at seed 0 (872 vs 876) and 3 at seed 1 (833 vs 836), under 0.5% of the pool. Second, at 10^{-5} and the step count of these MATH-500 runs the adapter changes the outputs little (Section E), so two nearly identical pools would give nearly identical models whatever the verifier did. We cannot separate these two explanations. The result is therefore: *removing one rare format defect had no detectable effect*, which the prevalence count alone predicts; it is not evidence about directional verifier bias in general, and it leaves untouched the gold-storage defect, which rejects 15.5–18.3% of correct in-loop solutions in both arms (Section B). Verifier-bias figures in the literature, such as the ∼14% of [6], are likewise conditional on a reference distribution; what matters for a loop is how often the affected forms occur in that loop's own output, which is cheap to count.

Pre-filter answer diversity. Over all problems of each run (in Design I the 500 selection and exam items together, not the exam alone) we also measured, before filtering, the fraction of problems on which all $k = 8$ samples give the same extracted answer, for Design P's 1.5B run and the 10^{-5} Design I runs at seed 0. It falls in every one: 24.8% → 1.7% for Design P's 1.5B run, 33.6% → 16.0% for Run C, 14.6% → 12.0% (A) and 14.6% → 12.2% (B). This agreement is mechanically tied to accuracy (problems on which every sample is correct are all-agreement problems) and to generations that hit the token cap, so it adds little beyond the accuracy change and does not by itself rule out stylistic convergence, which concerns how answers are written rather than which answer is given. The fraction of distinct answers among the k samples, which we had planned as the primary diversity measure, barely moves at 10^{-5} in the MATH-500 runs (0.542 → 0.554 for Run A, seed 0; 0.528 → 0.543 for Run B, seed 1) but rises for Run C (0.359 → 0.500) and in Design P (0.443 → 0.712). We report both for completeness and draw no conclusion from them.

Table S6. Runs A and B at 5×10^{-4} (MATH-500, $k = 4$, three rounds), held-out exam set of 150 problems. B differs from A only in the verifier patch; each seed draws its own exam partition. "Kept pool" is the number of training examples (= 2× optimizer steps). "Boxed" is the share of the 2,000 generations in each round containing `\boxed`.

Run, seed	Round	pass@1	pass@4	Kept pool	Boxed
A (orig.), 1	0	33.3%	50.0%	440	52%
	1	32.7%	49.3%	391	51%
	2	23.3%	42.7%	326	36%
B (patched), 1	0	33.3%	50.7%	443	52%
	1	35.3%	50.0%	392	49%
	2	28.7%	48.7%	328	41%
A (orig.), 2	0	31.3%	51.3%	424	54%
	1	31.3%	46.0%	386	50%
	2	27.3%	44.7%	350	37%
B (patched), 2	0	31.3%	52.0%	424	54%
	1	31.3%	44.7%	398	50%
	2	20.0%	40.7%	332	38%

G Higher Learning Rates in Full

Because 10^{-5} changed the MATH-500 outputs little, we reran the MATH-500 runs at higher rates with a smaller budget (three rounds, $k = 4$, so pass@4 replaces pass@8 and the pools are about half as large): Runs A and B at 5×10^{-4} at two seeds, and Run A at 10^{-4} at three seeds and at 2×10^{-4} at two. Seed 1 shares the exam partition of the seed-1 runs above and seeds 2 and 3 draw new ones. These runs are not directly comparable with the $k = 8$, five-round runs in metric or training dose. Within round 0 the loss falls from about 0.5 to 0.25 at 5×10^{-4} and from about 0.6 to 0.28 at 10^{-4} over about 220 steps.

At 5×10^{-4} every run declines, with wide intervals. By round 2 pass@1 has fallen 10.0 and 4.0 points (A) and 4.7 and 11.3 points (B), and pass@4 7.3, 6.7, 2.0 and 11.3 points (Table S6). The bootstrap intervals (Table 2) exclude zero for two of the four pass@1 changes and one (two more reach zero at their upper end) of the four pass@4 changes, and for the sample accuracy of all four. A and B at the same seed share the partition, the round-0 generations and almost the whole first pool, so the four runs are closer to two independent replicates than four. We read this as a consistent but imprecisely measured decline. The style shift does not separate imitation from drift. The verifier-accepted pool is 89–91% `\boxed` at round 0, far more than all generations (52–54%, Table S6), because unfinished generations are rarely boxed and rarely accepted. But the right comparison is with *finished* generations, and these start *more* boxed than the pool (95.6% and 98.5% on the exam at seeds 1 and 2; the pool also holds truncated and cap-hit examples that lose their final `\boxed`). Among finished exam generations the share falls to 69.7% (A, seed 1) and 82.9% (A, seed 2), from above the round-0 pool to below it, while the pool, drawn from these generations, falls too (73–77%

by round 2). A model imitating its less-boxed training data would move the same way, so this share cannot separate imitation of the pool from drift.

The decline may depend on the learning rate, on the evidence of one seed. Only seeds 1 and 2 were run at all three higher rates. At seed 1, sample accuracy at round 2 falls by 1.5, 3.5 and 8.3 points at 10^{-4}, 2×10^{-4} and 5×10^{-4}; at seed 2 by 3.7, 3.8 and 4.7, which is almost flat (Table 2; seed 3, run only at 10^{-4}, gives -1.0). The seed-1 series is monotone, the seed-2 series barely so, and at 10^{-4} and 2×10^{-4} the individual changes are about the size of re-sampling noise, so the rate dependence is suggestive, not established. The share of generations that hit the token cap does *not* follow the rate: it is flat or falls at 10^{-4} and 2×10^{-4} and rises only at 5×10^{-4} (Table 1).

The patch has no consistent effect. At round 2, B minus A is $+5.3$ [-2.0, $+12.0$] points of pass@1 at seed 1 and -7.3 [-14.0, -0.7] at seed 2. The sign flips with the seed, and the seed-2 interval excludes zero although the patch changes the first training pool by 0–3 examples. Same-seed runs that start from almost the same pool can thus end several points apart once the adapter moves (their later pools differ by 1–18 examples as the trajectories separate). This is one pass@1 interval whose upper end only just clears zero, so on its own it is weak evidence of trajectory noise; the replicate of Run A at seed 1 (Section H), which ends 3.2 points of sample accuracy and 4.1 points of matched finished accuracy away from the original, points the same way.

Two controls at 5×10^{-4}. The *no-filter* run trains on all k samples of every selection problem regardless of correctness (1,400 examples per round), removing the verifier from the loop. The *train-on-gold* run ignores the model's generations and trains on the 350 reference solutions, identical every round. Both were run at two seeds, on the partitions of Run A at the same seeds. Neither is a single-variable control: per round, no-filter takes 700 optimizer steps and train-on-gold 175, against 220, 196 and 163 for Run A at seed 1 (and 212, 193, 175 at seed 2). Train-on-gold is thus close to Run A in optimizer steps, so its comparison is mainly one of data; no-filter takes 3–4× as many steps, and since dose may matter (the rate dependence above is suggestive), dose and data differ at once in that comparison.

Removing the filter is never better than keeping it on any measure beyond noise, and is worse on pass@4 at both seeds, but the dose difference means this cannot be attributed to the filter. At round 2, no-filter minus Run A is $+4.0$ [-2.7, $+10.7$] (seed 1) and -4.7 [-12.0, $+2.7$] (seed 2) points of pass@1, of mixed sign, and -2.0 [-8.7, $+4.7$] and -5.3 [-12.0, $+1.3$] points of pass@4, negative at both seeds, so pass@4 weakly favours keeping the filter; for sample accuracy the difference is $+0.2$ [-3.3, $+3.7$] at seed 1 and -5.5 [-9.2, -1.8] at seed 2, so one single-seed interval excludes zero, in favour of the filter, and the other is centred on zero. The verifier gate keeps 30–31% of the samples at round 0 and 23–25% by round 2. With two seeds, a 3–4× larger training dose for no-filter, and intervals of this width, we can only say that effects of a few points in either direction are not excluded. Training on reference solutions is followed by a much larger decline: pass@1 falls to 18.0% and 15.3% (seed 1) and 8.7% and 10.0% (seed 2), and

pass@4 by about 21 points by round 2 at both seeds ($[-29.3, -12.7]$ at worst), while the model adopts the reference style (generations with a `$`-delimited span go from 5% to 98%, with `\boxed` from 52% to 86%). That accuracy is graded by the same verifier after a large format shift, so part of this drop could be verifier error. We checked this in two ways after the runs (post hoc). First, among the round-2 answers the verifier rejects, those whose final `\boxed` content equals the stored gold string after light normalisation are 14 of 600 generations at seed 1 and 11 of 600 at seed 2; counting them correct would raise round-2 sample accuracy by about 2 points. This is a lower bound: exact string match misses equivalent forms, and the about 14% of round-2 generations with no `\boxed` at all are not covered. Second, the gold-storage re-score of Section B makes the sample-accuracy decline larger, not smaller (-24.7 and -22.0 points against -19.7 and -18.7), because the defect penalises round 0 more; but it repairs one defect and does not test whether answers in the new style are parsed correctly. Neither check is a hand audit. Even several times the first bound would be far below a fall of about 19 points, and finished accuracy falls to 12–15%, so we think the fall is unlikely to be mainly verifier error. But both checks bound verifier error only from below, the unboxed generations are not checked at all, and we have no upper bound, so a format-shift error of unknown size remains. Its rounds are replicate fine-tunes on identical data (their final training losses agree to three decimals), so the difference between its rounds 1 and 2 mainly reflects generation sampling. No verifier-filtered self-training run at this rate shows a pass@k gain; their pass@4 falls by 2.0–11.3 points (no-filter 9.3 and 12.0), whereas train-on-gold falls by about 21.

Run C at 5×10^{-4}. We also reran Run C (GSM8K, seed 1, three rounds, $k = 4$) at 5×10^{-4}, once. pass@1 falls 70.7% $\rightarrow$ 64.7% $\rightarrow$ 48.7% (interval for the round-0-to-2 change $[-29.3, -15.3]$) and pass@4 86.0% $\rightarrow$ 86.0% $\rightarrow$ 76.7%. The cap-hit share rises from 3.8% to 15.0% and mean length from 958 to 1,313 characters, and accuracy on the finished generations falls clearly (71.9% $\rightarrow$ 63.3%; -12.1 points matched). In the three-part split of Section H, about half of this run's sample-accuracy decline (51%) is in the finished-accuracy part and 43% in the cap-share part (Table S9), so on GSM8K, in this single run, the loss is both less accurate solutions and more unfinished ones. This run uses a different partition, k and round count from the 10^{-5} Run C, so the two can be compared only through their within-run changes (-1.3 points of pass@1 by round 2 at 10^{-5}, -22.0 at 5×10^{-4}; the 10^{-5} run falls further later, to -12.7 by round 4). That is suggestive of a rate dependence on GSM8K too, from one run per rate.

H Anatomy of the Decline in Full

Everything in this subsection is post hoc: the cap-hit measure, the matched metric and the decomposition were defined after the runs, to explain the declines, and they describe the declines rather than identify causes. Sample-level accuracy mixes two things: whether a model that finishes a solution gets it

right, and whether it finishes within the 512-token cap at all. Token counts were not saved at generation time, so we re-tokenised the saved decoded text with the model's tokenizer and call a generation *cap-hit* if it has at least 508 tokens (`experiments/viec3/review_cpu/caphit.py`, `capanat.py`); the margin allows for small differences between generated and re-encoded token sequences. The threshold matters little: few generations fall near it (at most 37 of 600 per run and round have 480–507 tokens), a threshold of 500 raises the cap-hit share by at most about 2 points and one of 512 lowers it by at most about 4.0, and with either the share still rises only at 5×10^{-4}. The untrained model already hits the cap in 40.7–48.2% of its MATH-500 exam generations, depending on partition and sampling (43.0–43.2% on the $k = 4$ seed-1 and seed-2 partitions; 3.5–3.8% on GSM8K), and only 4–7% of those are accepted. The 512-token cap is therefore binding from the start, not a consequence of training.

An earlier proxy overstated termination. Earlier versions of this paper called a generation *degenerate* if it was longer than 1,500 characters and contained no `\boxed`, and read its rise (9.8% → 41.3% for Run A, seed 1) as a growing failure to finish. Measured against cap-hit, the proxy flags only 9.8% of round-0 generations when 43.2% hit the cap, and its rise mostly tracks a change in the text rather than in termination: proxy-flagged generations have 3.3 characters per token at round 0 and 4.9–5.0 at round 2, against 2.7–3.0 for the rest, so drifting prose crosses 1,500 characters within the same token budget. At round 0 every proxy-flagged generation is a cap-hit (at round 2, 80–98% are), but the proxy catches only 19–25% of cap-hits at round 0 and 58–75% at round 2 (at 5×10^{-4}), so its rise is largely a rise in the share of cap-hits it can see. At 10^{-4} and 2×10^{-4} the proxy share grows 2.0–2.6× (at seed 1, 9.8% → 19.5% at 10^{-4} and 9.8% → 25.0% at 2×10^{-4}) while cap-hit stays flat or falls.

At the highest rate, finished solutions are less accurate. In all six self-training runs at 5×10^{-4} (Table S7) mean length grows by 41–52%, but the cap-hit share rises by only 2.3–14.2 points, while accuracy on the generations that finish falls from 50.6–54.0% to 43.0–46.3%. Because the problems that still finish change, we compared last-round finished accuracy with the untrained model's finished accuracy on the *same problem mix*, weighting each problem's round-0 finished accuracy by the number of finished last-round samples it has. Problems with no finished last-round sample drop out (22–45 of the 150 per run); where a problem had no finished round-0 sample we use its accuracy over all round-0 samples (6–21 problems, 34–37 in the gold runs), which, being mostly cap-hit accuracy, biases the baseline down for those problems; intervals use 2,000 problem-level bootstrap resamples. Because training changes which problems finish, the matched metric conditions on an outcome of training and is descriptive, not a causal effect. On that matched basis finished generations lose 5.5–10.6 points, with intervals excluding zero in five of the six runs (Table 2). We also split the change in sample accuracy symmetrically into a finished-accuracy part $\bar{s}\,\Delta f$, a cap-share part $\Delta s\,(\bar{f} - \bar{c})$ and a cap-accuracy part $(1 - \bar{s})\,\Delta c$, where s is the share of finished generations, f and c the accuracy of finished and cap-hit generations, and bars denote round-0/last-round means. Only the cap-

Table S7. What the decline consists of (MATH-500, exam generations, round 0 → last round). "Cap-hit" is the share of generations whose decoded text re-tokenises to ≥ 508 tokens; "finished acc." is sample accuracy on the other generations, unadjusted; "matched Δ" is last-round finished accuracy minus the untrained model's finished accuracy on the same problem mix (intervals in Table 2). "Trunc." is the percentage of the round-0 training pool longer than the 512-token training limit. The first six rows are at 5×10^{-4}, then 2×10^{-4} and 10^{-4}; the 10^{-5} rows are the original runs ($k = 8$, round 0 → round 4, not comparable in dose); the last two rows (train on gold) are at 5×10^{-4}.

Run	Seed	Length (chars)	Cap-hit (%)	Fin. acc. (%)	Matched Δ	Trunc.
A (orig.)	1	1175 → 1678	43.2 → 46.2	54.0 → 43.0	−9.1	24.5
B (patched)	1	1175 → 1659	43.2 → 47.8	54.0 → 46.3	−6.8	24.6
A (orig.)	2	1154 → 1641	43.0 → 51.2	50.6 → 45.4	−5.5	24.5
B (patched)	2	1154 → 1728	43.0 → 54.0	51.2 → 44.2	−5.9	24.5
No filter	1	1175 → 1668	43.2 → 45.5	54.0 → 43.1	−10.6	62.4
No filter	2	1154 → 1755	43.0 → 57.2	50.6 → 43.6	−6.6	—
A, 2×10^{-4}	1	1175 → 1433	43.2 → 40.8	54.0 → 46.5	−4.0	24.5
A, 2×10^{-4}	2	1154 → 1431	43.0 → 39.7	50.6 → 41.7	−4.7	24.5
A, 10^{-4}	1	1175 → 1276	43.2 → 42.7	54.0 → 48.5	−4.3	24.5
A, 10^{-4}	2	1154 → 1284	43.0 → 42.8	50.6 → 44.6	−3.5	24.5
A, 10^{-4}	3	1178 → 1355	48.2 → 45.7	50.2 → 47.5	−0.6	—
A, 10^{-5}	0	1158 → 1147	46.9 → 47.4	54.8 → 51.0	−3.8	—
A, 10^{-5}	1	1158 → 1146	40.7 → 41.8	55.2 → 54.3	−2.3	—
Train on gold	1	1175 → 485	43.2 → 7.2	54.0 → 14.7	−25.6	17.4
Train on gold	2	1154 → 516	43.0 → 8.3	50.6 → 12.5	−25.9	—

share part measures termination; the cap-accuracy part is a change in how often unfinished text is still accepted. Table S9 gives the three parts and paired bootstrap intervals (2,000 resamples) for the cap-hit change and for the finished-accuracy share (`experiments/viec3/review_cpu/dot4/boot.py`). This decomposition uses the unmatched finished accuracy, so its finished part still contains the shift in which problems finish.

At 5×10^{-4} the cap-hit rise has an interval excluding zero in all three seed-2 runs and in B at seed 1; at seed 1 the other rises (+2.3 and +3.0) are about the size of two samplings of the untrained model on the same partition (43.2% at $k = 4$ against 40.7% at $k = 8$), and the replicate and the 1,024-limit run do not rise at all. The cap-share part is 12–30% of the decline in the three original seed-1 runs (−2% and 5% in the replicate and the 1,024-limit run) and 52–72% at seed 2; the cap-accuracy part is small (−1.1 to +1.5 points). The finished-accuracy shares have wide intervals, and they split by seed (seed-1 runs 62–89%, seed-2 runs 34–59%); since seed sets the partition, this may be partition variation rather than run-specific anatomy. What the table supports is qualitative: at 5×10^{-4} both parts contribute, in a balance that differs by partition. At 10^{-4} and 2×10^{-4} the cap-hit share does not rise, the finished-accuracy part exceeds the whole (smaller) change in four of the five runs and is 93% of it in

the fifth, and matched finished accuracy falls by 0.6 to 4.7 points with intervals including zero. Termination therefore contributes materially to the decline only at the highest rate, and clearly only at one seed (at 10^{-5} the declines are too small to divide reliably). The long generations that remain open as ordinary solutions and then drift into loosely related prose (e.g. "... acknowledged hurdles acknowledged setbacks ..."), with low exact repetition (mean word 5-gram repeat ratio about 0.02, similar to other generations, though the decoding repetition penalty of 1.1 suppresses exact repeats by itself); this description rests on inspection of examples, not on a systematic annotation. Train-on-gold is different: responses shorten by 55–59%, cap-hit falls from 43% to 7–8%, and finished accuracy collapses (-25.6 and -25.9 points matched). This collapse may owe something to the training recipe rather than to the data alone: the loss is taken over the prompt as well as the response, and reference solutions are short and terse, so even one epoch on 350 of them per round can pull the model far from its own reasoning style; we did not test a response-only loss.

Two checks of the pipeline. The training limit (`max_length`= 512 tokens, prompt included) equals the generation cap, so 24–26% of the kept pool (108 of 440 examples in Run A at round 0) is truncated in training: it loses its final answer and end-of-sequence token and could teach the model not to stop. We reran Run A at seed 1 and 5×10^{-4} once with a training limit of 1,024 tokens, together with a matched replicate that keeps the 512-token limit. Removing the truncation changed little in this one run: the round-2 cap-hit share is 43.8% with the longer limit, 43.0% in the replicate and 46.2% in the original Run A, and matched finished accuracy falls by 5.7, 5.0 and 9.1 points. The differences between the three arms are within the spread of the one same-configuration pair we have (the original and the replicate, which are themselves two of the three arms, 4.1 points apart matched), so this pair is a poor estimate of replicate noise; and the longer limit does not remove every unterminated example: 43 of the 440 accepted round-0 examples (9.8%) are cap-hit generations, which have no end-of-sequence token under any training limit. With one run per arm this can only exclude a truncation effect much larger than the 4-point gap between the pair. The replicate reproduces the direction of Run A's decline (pass@1 $33.3 \to 31.3 \to 25.3$%, pass@4 $50.0 \to 48.0 \to 44.0$%).

Re-scoring the same adapters under a 1,536-token cap, with everything else fixed (Table S8), is consistent with the 512-token cap not being the whole story, though with one adapter per row and no intervals it does not establish it. The untrained model gains *more* from the larger cap than the adapters do (sample accuracy +6.8 points, against +5.8, +2.0 and +3.5), so the adapters' deficit does not close: -8.0, -8.8 and -7.8 points at 512 tokens and -9.0, -13.7 and -11.2 at 1,536. This does not separate less accurate solutions from generations that do not stop: at 1,536 tokens the proxy's "Acc. finished" is close for all four rows (40.8–44.1%) while its "Degenerate" share is 2.5% for the untrained model and 23–33% for the adapters, which points to non-termination, though the proxy undercounts cap hits. Since re-scoring the same adapter moves pass@1 by up to 6.7 points, differences of one to three points between rows, including

Table S8. Adapters re-scored on the same exam problems (seed-1 partition, $k = 4$, 150 problems) under a 512- and a 1,536-token generation cap. Round-2 adapters from runs at 5×10^{-4} ("replicate" keeps the 512-token training limit, "1024 limit" trains with no truncated example) and 2×10^{-4}, against the untrained model. One adapter per row; re-scoring the same adapter moves pass@1 by 2.7–6.7 points against the run's own round-2 value. Only summary statistics were saved for these re-scorings, so no bootstrap intervals and no cap-hit counts are available; "Degenerate" and "Acc. finished" use the character proxy of Section H, which undercounts cap hits, and are shown only for completeness. Sample accuracy rests on 600 samples clustered in 150 problems.

Model	Cap	pass@1	pass@4	Sample acc.	Degenerate	Acc. finished
Untrained	512	36.0%	52.0%	36.3%	8.7%	39.2%
	1536	40.7%	59.3%	43.2%	2.5%	44.1%
5×10^{-4}, replicate	512	29.3%	46.7%	28.3%	30.0%	38.1%
	1536	35.3%	52.0%	34.2%	23.0%	43.3%
5×10^{-4}, 1024 limit	512	26.0%	45.3%	27.5%	37.2%	40.8%
	1536	30.7%	50.7%	29.5%	32.7%	41.3%
2×10^{-4}	512	27.3%	48.7%	28.5%	28.7%	37.9%
	1536	32.0%	52.7%	32.0%	23.7%	40.8%

the comparison of gains, are not interpretable. The data suggest that the gap persists at a tripled cap, whether from less accurate solutions or from generations that do not stop, for single adapters on one partition; they do not show what a still larger cap would do.

Table S9. Decomposition of the change in sample accuracy (points, round 0 $\rightarrow$ last round, MATH-500 exam unless stated) into finished-accuracy, cap-share and cap-accuracy parts, with the change in cap-hit share. Intervals: paired problem-level bootstrap, 2,000 resamples. The share is the finished part over the whole change.

Run, seed	Δcap-hit	Δacc.	Finished	Cap-share	Cap-acc.	Fin. share
5×10^{-4}						
A, 1	+3.0 [−1.7, 7.7]	−8.3	−6.0	−1.3	−1.0	73% [43, 108]
B, 1	+4.7 [0.2, 9.5]	−6.7	−4.2	−2.0	−0.5	62% [21, 100]
No filter, 1	+2.3 [−2.5, 6.8]	−8.2	−6.0	−1.0	−1.1	74% [46, 106]
A, 2	+8.2 [3.8, 12.3]	−4.7	−2.7	−3.4	+1.5	59% [−11, 114]
B, 2	+11.0 [6.2, 16.0]	−9.2	−3.6	−4.8	−0.8	39% [6, 64]
No filter, 2	+14.2 [9.3, 18.8]	−10.2	−3.5	−6.1	−0.6	34% [5, 57]
A replicate, 1	−0.2 [−4.5, 4.3]	−5.2	−4.6	+0.1	−0.7	89% [41, 162]
A, 1024 limit, 1	+0.7 [−3.7, 5.2]	−5.7	−4.8	−0.3	−0.5	85% [42, 169]
C (GSM8K), 1	+11.2 [7.8, 14.7]	−15.3	−7.8	−6.6	−0.9	51% [30, 66]
2×10^{-4} *and* 10^{-4}						
A, 2×10^{-4}, 1	−2.3 [−6.7, 2.3]	−3.5	−4.3	+1.0	−0.2	124%
A, 2×10^{-4}, 2	−3.3 [−7.7, 1.0]	−3.7	−5.2	+1.4	+0.1	142%
A, 10^{-4}, 1	−0.5 [−4.5, 3.7]	−1.8	−3.1	+0.2	+1.0	169%
A, 10^{-4}, 2	−0.2 [−4.3, 4.0]	−3.7	−3.4	+0.1	−0.3	93%
A, 10^{-4}, 3	−2.5 [−6.3, 1.3]	−1.0	−1.4	+1.1	−0.7	139%

I Discussion in Full

I.1 Answer to the Research Question

Does the model get better at mathematics? Not detectably. On held-out problems no run raises pass@1 or pass@k by an amount whose interval excludes zero, at any of four learning rates, and the sample-accuracy intervals exclude gains larger than about 2.2 points. This bound is loose on three counts: it is read off the observed intervals, not a pre-specified equivalence test; it uses sample accuracy, which we chose as primary after the runs (on the pre-set pass@1 and pass@k the upper ends reach +8.0 and +7.3); and the intervals omit training-seed variance, which is of the same size (a replicate of Run A at 5×10^{-4}, seed 1, ends 3.2 points of sample accuracy away). In every MATH-500 run except 10^{-4} seed 3 (16/14) the problems lost over the rounds equal or outnumber the problems gained (Table 1; 10^{-5} runs: 20/21 to 32/34 over four transitions), which restates the sign of the pass@k change rather than adding evidence. Most per-problem movement is churn: about 2 to 25 problems are gained and about as many lost per round even where the averages barely move.

Does it learn to write what the verifier accepts? The data cannot settle this. The pattern fixed in advance for this reading (pass@1 up, pass@k flat, style converging) requires pass@1 to rise, so it cannot fire in runs where accuracy falls, and pass@1 never rises detectably (the largest rise is +0.7 [−6.7, +8.0], at 10^{-4}, seed 1); the rule is one-sided and its silence is not evidence against verifier-style learning. Its style leg, the pre-set `\boxed` share, was evaluated on its own and also fails: in self-training the share does not converge but falls (all 2,000 generations per round: 52% → 36% and 54% → 37% at 5×10^{-4}, Table S6; the 600 exam generations: 55.5% → 38.3% and 58.5% → 42.5%). The post hoc refinements do not settle it either. The kept pool (89–91% boxed at round 0) is less boxed than the *finished* exam generations it is compared with (95.6% and 98.5%), and the finished share falls (69.7% and 82.9%) as the pool's does (73–77% by round 2). That is what imitation of the pool would predict as well as what drift would, so the share does not discriminate. The post hoc false-reject audit (Section 5.3) finds no adaptation to the gold-storage defect once the falling `\boxed` share is accounted for, but it has little power. The `\boxed` share is also not a clean measure of what the verifier prefers, because the gold-storage defect (Section B) penalises some `\boxed` answers. The one strong style convergence (`$` spans 5% → 98%, `\boxed` 52% → 86%) happens under train-on-gold, with no verifier in the loop and with falling accuracy: style converging alone, which by our rule says nothing about verifier-style learning.

What does change? In this post hoc description, most clearly at the highest rate, the model gets less accurate on the solutions it does finish. The matched intervals exclude zero in five of six MATH-500 runs at 5×10^{-4}, in both GSM8K runs (10^{-5} and 5×10^{-4}) and, on MATH-500 at 10^{-5}, only just, for A and B at seed 0, which are close to one replicate; at 10^{-4} and 2×10^{-4} the point estimates are negative but the intervals include zero (both GSM8K figures are single runs). More generations run into the token cap only at 5×10^{-4}: by 2–14 points in the

four verifier-filtered runs and two no-filter controls on MATH-500, with intervals excluding zero at seed 2 and for one run at seed 1, and not in a replicate; the change in cap-hit share accounts for 12–30% of the decline in the original seed-1 runs and 52–72% at seed 2 (Table S9). At lower rates the cap-hit share changes by −3.3 to +1.1 points. The untrained model already hits the cap in 41–48% of MATH-500 generations, so the cap limits every round.

What happened to the pooled collapse? It does not reproduce when each dataset gets its own adapter and is scored on held-out problems. Because the pooled design also trained on about 4.5× more examples per round, and our higher-rate runs suggest (on one seed) that stronger training produces larger declines, which are at least compatible with a dose effect (we did not apply the measures of Section H to Design P, and its acceleration under shrinking pools is not explained by dose, Section D), we cannot say whether pooling, selection-set scoring or training dose produced it. We no longer claim that pooling and selection-set scoring "manufactured" the collapse.

I.2 Interpretation

An earlier version of this work explained the Design P curves with a self-reinforcing shrinking-pool mechanism: the biased verifier accepts a smaller, more style-skewed set each round and the skew compounds. The data give it no support, but they do not rule it out either. The patch removes a severe but rare bias and changes the pool by a handful of examples, so it tests how strong the bias is per answer, not whether a frequent bias compounds; the no-filter control declines without any verifier, as much as Run A at seed 1 and more at seed 2 (sample accuracy −10.2 against −4.7), but it also trains on 3–4× more examples, so its decline is confounded with dose; and the training-dose difference between the designs offers a simpler account that we have not tested directly either. Pool size is also partly definitional (it is sample accuracy on the selection items times the number of samples), so its co-movement with accuracy, which holds on pass@1 in Design I as well (Run C: pool −20%, pass@1 −12.7; Run A: −3%, −2.7), says little about mechanism.

Design I differs from Design P in at least four respects at once (isolation, a held-out split, n per run falling from 1,000 pooled problems to 500, and the training dose), so the P/I contrast is descriptive. The clean comparisons are internal to Design I and its higher-rate counterparts, and even there the controls differ in dose. The defensible statement about the verifier is narrow: *removing a format defect present in under* 1% *of generations, or removing the filter entirely, made no consistent difference at the resolution of this setup.* The one frequent systematic false negative we measured, the gold-storage defect, was present in every run and was never varied, so whether systematic false negatives are harmless in general, as Egashira et al. find for RLVR, our data cannot say.

I.3 Practical Suggestions

These suggestions are not findings of this paper: none was tested here. They are general good practice that problems in our own runs (one 1.5B model, one rule-based verifier, LoRA SFT, at most three seeds per condition) illustrate, and our data do not show that following them changes a conclusion. *Hold out evaluation items and keep one adapter per dataset* if you report per-dataset trajectories, so the curves can at least be read; our data do not rank these factors. *Match training dose across arms*, or report optimizer steps: pool size changes with filtering, k and dataset, and with one epoch it sets the number of updates. *Check that the training step changes the model* (loss and output statistics after one round) before reading a flat curve as stability. *Count the prevalence of a suspected verifier defect in the loop's own outputs before building a control around it;* a treatment that changes 0.5% of the pool cannot be detected with 150 problems. *Save token counts and report cap-hit directly,* not a character-length proxy, and adjust any "accuracy on finished solutions" for which problems still finish; our proxy misread a change in text style as a change in termination. *Report intervals,* and track pass@k and per-problem gains and losses alongside pass@1.

J Limitations in Full

1. *Unequal training doses.* With one epoch and batch size 2, optimizer steps are half the pool size: round 0 takes about 1,981 steps (inferred) in Design P, 436 in Run A and 951 in Run C at 10^{-5}, 220 in Run A at 5×10^{-4}, 700 for no-filter and 175 for train-on-gold. Every cross-arm comparison mixes the intended variable with dose, and the cause of the Design P decline is not identified.
2. *Underpowered controls.* The patch changes 0–4 training examples; at 10^{-5} the adapter also changes the outputs little. Its null is expected whatever the truth, and we report no equivalence bound. The no-filter and train-on-gold controls have two seeds each.
3. *Intervals cover only part of the uncertainty.* The bootstrap intervals cover problem draw and generation sampling, not training-seed variance. Most conditions have two or three seeds; the following have one: Run C at both rates, the truncation check (one replicate pair at 5×10^{-4} and one run with a 1024-token training limit), and each token-cap re-scoring. pass@1 is a single draw per problem.
4. *Learning rate.* The original 10^{-5} was an untuned default; the evidence that it changes outputs little is behavioural (unchanged length and cap-hit share), since its loss does fall. The higher-rate grid is coarse and uses three rounds and $k = 4$.
5. *Cap-hit is reconstructed:* token counts were not saved, so we re-tokenise the decoded text and count a generation as a cap hit at ≥ 508 of 512 tokens. The token-cap re-scorings saved only summary statistics, so their termination figures still rest on the character proxy, which we show undercounts cap hits.

6. *Verifier measurements.* The false-negative audit used other models' solutions and a scripted correctness check, inspected only rejections (so no false-positive rate). The gold-storage defect was measured afterwards on the loop's own outputs by re-scoring (Section B); it lowers absolute pass@1 by 4.7–8.0 points in the three self-training runs we re-scored (other runs are extrapolation), was present in every run including both arms of the patch control, and was never varied in a training run. It overlaps with the unparseable-gold correction (46 of the 51 unparseable golds become parseable once boxed), so the two corrections must not be added. The re-scored false-negative share assumes the re-scored verifier makes no false accepts.
7. *Restart-from-base design and scope.* A fresh adapter on the base model each round leaves compounding little room, so the absence of compounding is partly a property of the design. One model size in Design I, one verifier, one training method, at most five rounds; full fine-tuning, continued training and online RL are untested.
8. *The selecting verifier also grades.* The same math-verify configuration that chooses the training data scores the exam, so any systematic verifier error enters both selection and measurement. The gold-storage re-score (Section B) and the string audit of the train-on-gold runs bound part of this, but only for five runs and one defect class.
9. *A binding token cap.* At 512 tokens the untrained model already fails to finish 41–48% of MATH-500 generations (depending on the seed's exam split), so every measure is taken in a regime where length and accuracy are entangled, and the 512-token training limit truncates about a quarter of the kept pool. Our conclusions are about this regime; a larger cap during the loop is untested.
10. *No equivalence margin and many intervals.* We did not fix in advance how small a gain would count as "no gain", so the 2.2-point bound is read off the observed intervals. Table 2 alone holds 76 intervals and the text reports about ten more, none with a multiplicity correction. They are not independent (they share runs and problems), so "about four would exclude zero by chance" is only a rough guide, but single intervals that only just exclude zero should be read cautiously.
11. *Possible test-set contamination.* MATH-500 and GSM8K are public and may be in Qwen2.5's training data. This would raise absolute accuracy, and it need not affect all arms or rounds alike: if training on self-generated solutions makes the model forget memorised items, part of a within-run decline could be loss of memorisation rather than of skill. We did not test it.
12. *Training recipe.* The loss covers the prompt as well as the response, and we tried one LoRA configuration (rank 16 on query and value projections). Both could matter, most plausibly for the train-on-gold collapse, and neither was varied. No arm in any design improves detectably on its round 0, including training on reference solutions, so we have no positive reference showing that this recipe can help a 1.5B model at all; the declines may partly reflect the recipe rather than self-training.

13. *Limited power.* With 150 exam problems the smallest changes the design would detect with 80% power are about 10 points of pass@1, 8–10 of pass@k and 3.5–5.5 of sample accuracy (Section 3.4).
14. *A weak adaptation audit.* The false-reject audit of Section 5.3 was defined after the runs, covers one defect class whose effect is bound to 54–62 problems (only 24–28 of them escapable by any answer form), rests on 2–12 training examples per round that could teach the avoidance, and its two injected-bias runs, neither a valid positive control (a hash-chosen 30% injected `\boxed` bias and a bias on fraction golds, one seed each, Section 4.6 of the main paper) showed no selective avoidance, so we do not know it would detect adaptation.
15. *Confounded learning-rate grid and one model family.* The higher rates also use $k = 4$, three rounds and smaller pools, so only comparisons among them isolate the rate. All runs use Qwen2.5 models, whose behaviour under RL with spurious rewards is known to differ from other families [9].

K Future Work

Separate the explanations of the pooled decline with a pooled run scored on a held-out split and an isolated run matched in optimizer steps (by fixing the step count per round rather than the number of epochs); save token counts at generation time and test a larger generation cap during the loop, since the untrained model already hits 512 tokens in 41–48% of MATH-500 generations; refine the learning-rate grid and replicate Run C; and run a genuine positive control (a verifier biased against a frequent format) to show what effect size this design can detect. For the false-reject audit, a positive control would inject a defect that rejects a frequent, avoidable answer form (for example all `\boxed` answers on a fixed tenth of the problems) and check that r_t on those problems falls; without it a flat r_t cannot be read as absence of adaptation.

The gold-storage defect (Section B) is the natural next verifier control. We implemented and unit-tested a patch that wraps the gold in `\boxed{}` whenever the prediction contains one and the gold does not; re-scoring shows that in a training run it would enlarge the round-0 kept pool by about a fifth (440 → 542 and 872 → 1,066), roughly 34–49 times the change made by the `\dfrac` patch, so unlike Run B such a run would actually perturb the training data. Because a larger pool also means more optimizer steps, the comparison should hold dose fixed, either by subsampling the patched pool to the unpatched size or by fixing the step count. Given that same-configuration runs end several points apart, it needs several seeds per arm; a single seed would detect only a large effect.

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